

Risk Across Tails and Timescales

Does crash co-movement depend on the investment horizon, and what does that do to multi-day Expected Shortfall? A wavelet-copula study with out-of-sample evidence.

Equal-weight US sector ETF portfolio: XLE, XLF, XLK, XLV, XLI, XLU, SPY
Daily data 1999-01-05 to 2026-09-30 (6,977 returns); out-of-sample 1,695 days from 2020-01-01

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Answer in one paragraph. Model-free lower-tail dependence of the seven sector residuals is essentially flat across wavelet horizons ($\lambda_L(5\%) = 0.40$ at 2-8 days versus 0.36 beyond 32 days; paired bootstrap $p = 0.31$), so in normal times horizon-specific copulas add nothing measurable out of sample. What matters is the **tail shape**, not the scale: Gaussian-copula ES is rejected by the McNeil-Frey test at 3 of 3 levels, while fat-tailed copulas are not. In crises, long-horizon tail dependence jumps to 0.92 (calm: 0.39). For 20-day ES, the square-root-of-time shortcut is about 17% higher than a path-simulated ES, and the lower path ES is the one the data reject. Our recommendation is to keep sqrt-time scaling as a floor, move the daily engine to a Student-t copula, and add a crisis-dependence stress overlay.

1. Question, motivation and contribution

A risk desk usually estimates one dependence structure on daily data and then scales it to the holding period. The Basel FRTB framework does the same: ES at 97.5% is computed on a 10-day base horizon and stretched to longer liquidity horizons with a square-root-of-time rule. Both practices assume that the way assets crash *together* is the same at every horizon. The wavelet-copula literature suggests otherwise: dependence, and in particular lower-tail dependence, can differ between short and long time scales, and can shift in crises. We ask two linked questions for a liquid US sector portfolio: **(Q1) does lower-tail dependence change with the investment horizon, and (Q2) does ignoring that change mis-state 1-day, 5-day and 20-day VaR and ES?**

Published wavelet-copula studies mostly report in-sample, scale-by-scale copula parameters. Our contribution is to test the claim the way a model-validation team would. (i) We check the parametric tail estimates against a **model-free** estimator with block-bootstrap confidence intervals and a formal test of “tail dependence differs by scale”. (ii) We run a full **ablation** of 8 risk models out of sample, so that the effect of the wavelet step is separated from the effect of the copula family. (iii) We measure what horizon dependence does to **multi-day ES**, the quantity a regulator actually charges capital on, against the sqrt-time shortcut. (iv) We score ES with the strictly consistent FZ0 loss and Diebold-Mariano tests, not only with exceedance counts.

2. Data

We use daily adjusted closes from Yahoo Finance for six SPDR sector funds (energy XLE, financials XLF, technology XLK, health care XLV, industrials XLI, utilities XLU) and the S&P 500 fund SPY, 1999-01-05 to 2026-09-30. The end date is fixed in the configuration and the downloaded prices are cached in the repository, so every run sees identical data. Log returns are taken on the common trading calendar (6,977 days). The portfolio is equally weighted and its return is the cross-sectional mean of log returns. The sample covers the dot-com bust, the global financial crisis, the 2011 euro crisis, COVID-19 and the 2022 rate shock. Estimation uses data up to 2019-12-31; everything after is strictly out of sample.

3. Method

3.1 Margins

Each asset follows an AR(0)–GARCH(1,1) with standardised Student-t innovations, $r_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$. In-sample persistence $\alpha + \beta$ ranges from 0.987 to 0.999 and the t degrees of freedom from 6.2 to 11.7, so returns are strongly heteroskedastic and fat-tailed. Out of sample, parameters are re-estimated every 20 days on a rolling 1,000-day window, but σ_t is **updated every day** with the one-step GARCH filter, so forecasts react to news on the day it arrives. Standardised residuals are mapped to pseudo-observations by ranks, $u = \text{rank}/(n+1)$.

3.2 Horizon decomposition with the MODWT

We apply the maximal-overlap discrete wavelet transform (Daubechies db2, $J = 6$) to each residual series and use its multiresolution analysis (MRA), $z = D_1 + \dots + D_J + S_J$. Detail D_j captures movements at periods of 2^j to 2^{j+1} trading days. The MRA is computed exactly in the frequency domain as $D_j = \text{IDFT}(|H_j(f)|^2 X(f))$, which is zero-phase and reconstructs z to machine precision (a unit test checks the error is below 10^{-10}). Series are reflected at the ends, and each band is trimmed by the level- j boundary length $L_j = (2^j - 1)(L - 1) + 1$ at both ends so no boundary-contaminated value enters a copula fit. We group details into three horizon bands: **H1** = $D_1 + D_2$ (2–8 days, roughly one week), **H2** = $D_3 + D_4$ (8–32 days, roughly one month) and **H3** = $D_5 + D_6 + S_6$ (more than 32 days).

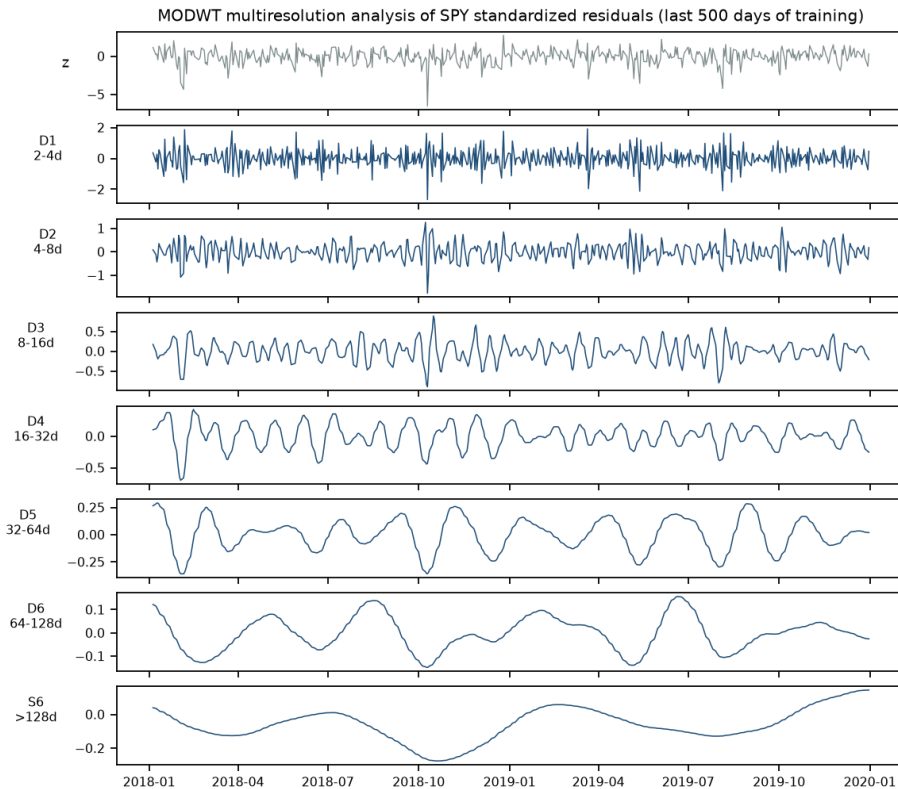


Figure 1. MODWT multiresolution analysis of SPY standardised residuals over the last 500 training days. Components sum exactly to the top panel; each lower panel isolates one band of periods.

3.3 Copulas and tail dependence

On each band (and on the raw residuals) we fit three seven-dimensional copulas by full maximum likelihood, so AIC values are directly comparable: the **Gaussian** copula (no tail dependence), the **Student-t** copula (symmetric tail dependence, correlation from Kendall's τ via $\rho = \sin(\pi\tau/2)$, ν by profile likelihood) and the **Clayton** copula (lower-tail dependence only, one parameter θ). The implied lower-tail coefficients are $\lambda_L = 2 \cdot t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$ for the t copula, averaged over the 21 pairs, and $\lambda_L = 2^{-1/\theta}$ for Clayton. Because each family forces a tail shape, we also estimate tail dependence **without a model**: $\lambda_L(q) = P(U_i < q, U_j < q)/q$, averaged over pairs, at $q = 5\%$ and 10% . Confidence intervals and the test of H3 against H1 use the Politis-Romano stationary bootstrap (mean block 20 days, 500 replications), resampling the same time indices in every band so the H3–H1 difference is paired.

3.4 Out-of-sample risk engine and ablation

From 2020-01-01 we refit margins and copulas every 20 days (85 refits). At each refit we simulate 20,000 joint shocks from the copula, transform them to t margins and, on each subsequent day, rescale them by that day's filtered σ to obtain the portfolio return distribution and its 1-day VaR and ES at 95%, 97.5% and 99%. The ablation crosses **dependence input** (raw daily residuals versus the H1 band) with **copula family** (Gaussian, t, Clayton), and adds two industry baselines: filtered historical simulation (FHS, the empirical residual vector rescaled by today's σ) and plain historical simulation (HS, last 500 days). All models share the same margins and random numbers.

3.5 Multi-day ES

For $h = 5$ and 20 days we compare four ways of producing h-day ES: (a) **sqrt-time Gaussian**, the 1-day Gaussian-copula ES times $\sqrt{h}$, which is the regulatory shortcut; (b) **sqrt-time t**, the same with the t copula; (c) **daily t path-sum**, which simulates h daily t-copula shocks along the GARCH expected-variance path and sums them; and (d) **horizon-matched**, which builds each asset's h-day distribution from the same paths but re-couples assets with the copula fitted on the matching band (H1 for 5 days, H2 for 20 days). Forecasts are evaluated on non-overlapping h-day portfolio returns (339 five-day and 84 twenty-day blocks).

3.6 Backtests and model comparison

VaR coverage uses the Kupiec unconditional-coverage test and the Christoffersen independence and conditional-coverage tests. ES uses the McNeil-Frey one-sided bootstrap test on exceedance residuals (loss/ES – 1), where a small p-value means ES is too low. Because exceedance counts cannot rank ES forecasts, we also score each model with the FZ0 loss of Patton, Ziegel and Chen (2019), which is strictly consistent for the (VaR, ES) pair, and compare models with Diebold–Mariano tests using Newey–West standard errors. Lower FZ0 is better.

4. Results: does tail dependence change with horizon?

Band	n	AIC Gaussian	AIC t	AIC Clayton	ν (t)	λ_L t	λ_L Clayton	λ_L (5%) model-free [95% CI]
daily	5,282	-30,796	-32,804	-16,910	7.2	0.223	0.421	0.428 [0.398, 0.464]
H1 2-8d	5,262	-30,417	-32,291	-16,486	7.7	0.215	0.411	0.398 [0.366, 0.440]
H2 8-32d	5,190	-34,072	-35,636	-16,976	8.6	0.200	0.426	0.392 [0.351, 0.438]
H3 >32d	4,902	-29,666	-29,968	-14,809	15.5	0.090	0.408	0.365 [0.299, 0.444]

Table 1. In-sample (to 2019) copula fits by horizon band. Bold marks the AIC winner. λ_L columns compare two parametric tail coefficients with the model-free estimate.

The Student-t copula wins on AIC in every band, and it also wins in every one of the 85 out-of-sample refits for the raw, H1 and H2 inputs. Clayton is far behind: a single θ cannot fit 21 different pairwise correlations, so its likelihood collapses. This reverses the result of our own first draft, which used a pairwise composite likelihood for Clayton whose AIC was not comparable with the full likelihoods of the other families.

Parametric and model-free answers disagree. Read through the t copula, tail dependence falls with horizon, from 0.22 at H1 to 0.09 at H3, because the fitted ν rises from 7.7 to 15.5. The model-free estimate does not show this: it moves from 0.398 to 0.365, the paired difference is -0.033 with 95% interval [-0.103, +0.041] and $p = 0.31$. At $q = 10\%$ the difference is -0.011 ($p = 0.67$). The parametric decline is driven by the t copula tying tail dependence to the body of the distribution: smoother long-scale components look more Gaussian in the body, which pushes ν up and mechanically shrinks λ_L , even though joint extremes are as frequent as before. Scale-wise tail coefficients reported from a fitted family should therefore be cross-checked against a model-free estimate.

A second check aggregates residuals into non-overlapping 5- and 20-day sums instead of filtering them. $\lambda_L(5\%)$ is 0.424 for daily, 0.432 for 5-day and 0.361 for 20-day sums, with the 20-day interval [0.26, 0.49] covering both other estimates. Two different horizon constructions give the same verdict: **in the full sample, crash co-movement is statistically indistinguishable across horizons.**

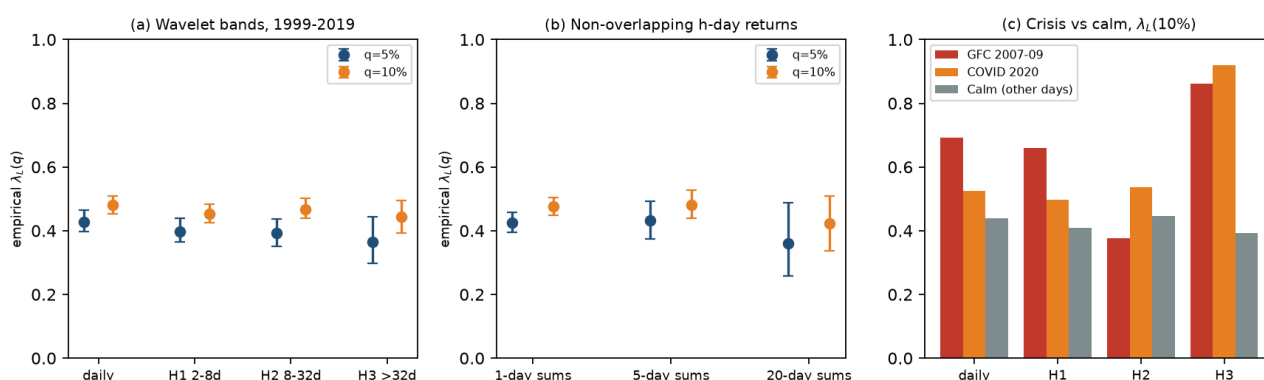


Figure 2. Model-free lower-tail dependence with 95% stationary-bootstrap intervals. (a) by wavelet band; (b) by non-overlapping aggregation horizon; (c) crisis windows against all other days at $q = 10\%$.

Window	Days	daily	H1	H2	H3
GFC 2007-09	504	0.69	0.66	0.38	0.86
COVID 2020	232	0.53	0.50	0.54	0.92
Calm (other days)	6,241	0.44	0.41	0.45	0.39

Table 2. Model-free $\lambda_L(10\%)$ in crisis windows against calm days, full sample.

Crises change the picture. Table 2 splits the full sample into the global financial crisis, the COVID crash and all remaining days. In calm periods tail dependence is again flat across bands (0.41 at H1, 0.39 at H3). In both crises the long-horizon band jumps to 0.86, 0.92: when a crisis lasts months, the slow components of all sectors fall together. This differs in emphasis from Alqaralleh and Canepa (2021), who find COVID raised dependence mainly at 2–8 day scales for national indices. We add a caveat: the H3 band is very persistent, so a crisis window of a few hundred days holds only a handful of independent long-scale moves, and these crisis numbers are descriptive rather than tested.

5. Results: 1-day VaR and ES out of sample

Model	Hit 95%	Hit 97.5%	Hit 99%	Kupiec p 97.5%	CC p 97.5%	MF p 95%	MF p 97.5%	MF p 99%	Loss/ES 97.5%	Mean ES 97.5%	FZ0 97.5%
H1-student	5.84%	2.89%	1.36%	0.314	0.109	0.107	0.070	0.091	1.082	2.55%	-3.6110
FHS	4.31%	1.42%	0.71%	0.002	0.001	0.806	0.069	0.118	1.039	3.06%	-3.6081
raw-student	5.72%	3.07%	1.30%	0.148	0.028	0.073	0.116	0.051	1.069	2.55%	-3.6079
H1-clayton	5.66%	2.36%	1.24%	0.709	0.581	0.546	0.155	0.408	1.050	2.78%	-3.6077
raw-clayton	5.66%	2.30%	1.18%	0.595	0.510	0.610	0.164	0.367	1.049	2.80%	-3.6058
H1-gaussian	5.78%	3.07%	1.36%	0.148	0.028	0.046	0.046	0.034	1.097	2.48%	-3.6045
raw-gaussian	5.84%	3.07%	1.42%	0.148	0.028	0.043	0.036	0.020	1.106	2.46%	-3.6029
HS	4.25%	2.42%	1.47%	0.830	0.002	0.023	0.011	0.043	1.217	3.72%	-3.1789

Table 3. Ablation of 8 models over 1,695 out-of-sample days, sorted by FZ0 loss at 97.5% (highlighted row is best). Expected hit rates are 5%, 2.5% and 1%. MF is the McNeil-Frey test, where a small p-value means ES is too low. Loss/ES is the mean realised loss on exceedance days divided by the mean forecast ES on those days.

Comparison (A vs B)	DM 95%	p	DM 97.5%	p	DM 99%	p
raw-student vs raw-gaussian	-0.46	0.646	-0.95	0.342	-1.55	0.121
raw-clayton vs raw-gaussian	-0.35	0.723	-0.17	0.864	-0.84	0.403
H1-gaussian vs raw-gaussian	-0.71	0.479	-0.41	0.682	-1.54	0.122
H1-student vs raw-student	-0.38	0.702	-0.87	0.383	-0.99	0.322
H1-clayton vs raw-clayton	+0.43	0.669	-0.49	0.623	-0.65	0.517
raw-student vs FHS	+0.20	0.842	+0.01	0.995	+0.57	0.571
HS vs raw-gaussian	+2.89	0.004	+2.89	0.004	+2.67	0.007

Table 4. Diebold–Mariano tests on FZ0 loss. A negative statistic means model A has lower loss than model B.

The wavelet step does not improve 1-day forecasts. Replacing raw residuals with the H1 band changes FZ0 loss by amounts that no Diebold–Mariano test detects (smallest p-value 0.12). H1-student has the lowest FZ0 at 97.5%, but its edge over the plain t copula is not significant ($p = 0.38$). The scale result of Section 4 predicts exactly this: if tail dependence is flat across horizons, filtering out long-scale components removes little information about tomorrow's crash.

The copula family matters for ES. The Gaussian copula's ES is rejected by McNeil–Frey at 95%, 97.5%, 99%: on exceedance days losses are 11% larger than its ES at 97.5%. The t copula raises mean ES by 3.7% and is not rejected at 97.5% ($p = 0.12$). Clayton raises it by 14.0% and is the only parametric family whose loss/ES ratio is close to one at all levels and which passes every coverage and ES test, despite having by far the worst AIC. This is the central lesson of the ablation: **AIC rewards fit in the body of the distribution, while risk lives in the lower tail**. Model choice for ES should be made on tail scores such as FZ0 and McNeil–Frey, not on likelihood alone. FZ0 does not separate the copula models significantly (Table 4), because the GARCH volatility filter, which all of them share, does most of the work. That shared filter is also why every copula model beats plain HS by a wide, significant margin (DM $p = 0.004$), and why FHS is close to the copula models.

Exceedances still cluster. Christoffersen conditional coverage is rejected at 97.5% for 5 of 8 models (raw-gaussian, raw-student, H1-gaussian, FHS, HS), mostly because of March 2020 (Figure 3), when volatility rose faster than any GARCH filter could follow. Static copulas cannot fix this; Section 8 discusses what would.

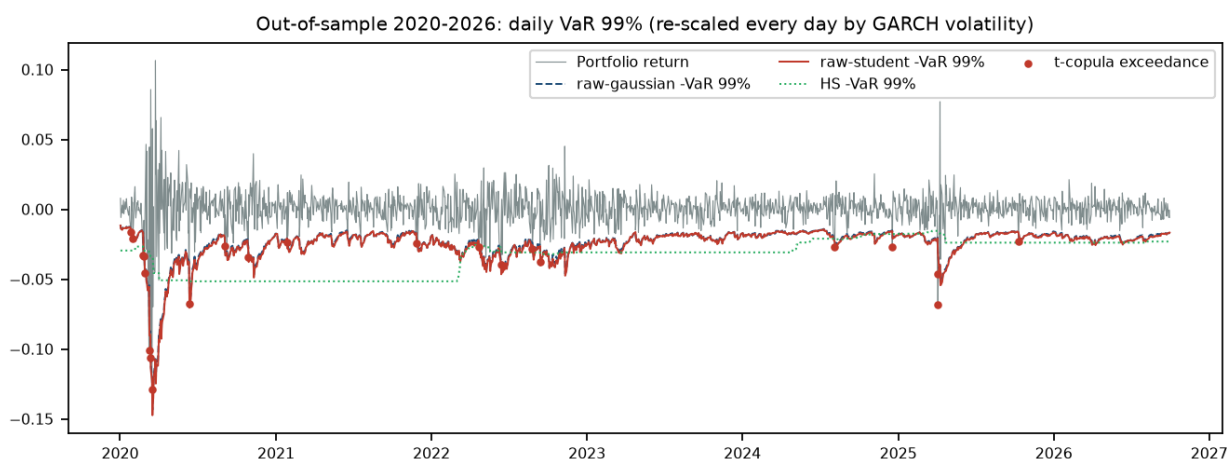


Figure 3. Out-of-sample portfolio returns with 99% VaR from three engines. Copula VaR moves daily with the GARCH filter; HS VaR adjusts only as its window rolls.

5.1 Where the differences come from: sub-periods

Period	Days	Expected hits	raw-gaussian	raw-student	raw-clayton	H1-student	FHS	HS
2020 (COVID)	253	6.3	15 0.009 1.21	15 0.021 1.16	15 0.114 1.08	15 0.021 1.16	11 0.295 0.98	16 0.029 1.31
2021-2022 (rate shock)	503	12.6	20 0.710 0.96	19 0.896 0.93	16 0.998 0.89	19 0.881 0.94	7 0.911 0.91	11 0.129 1.06
2023-2026 (calm)	939	23.5	17 0.154 1.16	18 0.239 1.11	8 0.055 1.34	15 0.138 1.17	6 0.006 1.51	14 0.139 1.18

Table 4b. 97.5% backtests by sub-period. Each cell shows exceedances | McNeil-Frey p | loss/ES on exceedance days.

Splitting the out-of-sample period shows that the ES differences are a crisis story. In 2020 (COVID), McNeil-Frey rejects raw-gaussian, raw-student, H1-student, HS, and only raw-clayton and FHS survive. In 2021-2022 (rate shock), no model is rejected, including the Gaussian copula. In 2023-2026 (calm) Clayton becomes too conservative: it records 8 exceedances against 23.5 expected (Kupiec $p < 0.001$). Figure 4b shows the mechanism. Relative to the Gaussian copula, the t copula's ES add-on is small and stable (median 3.5%, 10-90% range 1.1% to 6.6%), while Clayton's is a near-constant 14%. A fixed tail add-on is right in a crash and too much in calm markets, which is why we recommend Clayton as a challenger number rather than the production engine. Figure 4a confirms that the t copula's AIC advantage over the Gaussian holds at every refit and for every dependence input, so the family ranking is not a product of one period.

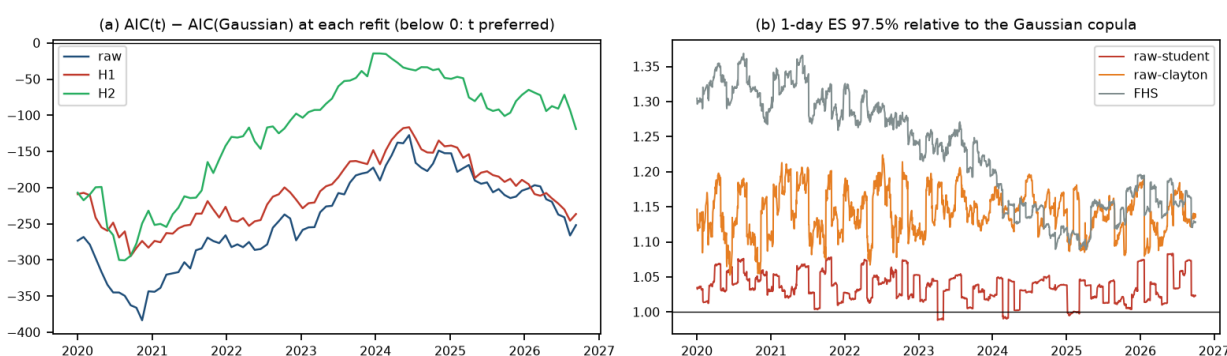


Figure 4. (a) AIC difference between t and Gaussian copulas at each of the out-of-sample refits, by dependence input. (b) Daily 97.5% ES of three models divided by the Gaussian-copula ES.

6. Results: multi-day ES and the square-root-of-time rule

h	Method	Mean ES 95%	Mean ES 97.5%	Mean ES 99%	ES ratio vs sqrt-G 97.5%	Hits 97.5% (exp.)	MF p 97.5%	Loss/ES 97.5%	FZ0 97.5%	DM p vs sqrt-G
5	sqrt-time Gaussian	4.61%	5.46%	6.60%	1.000	14 (8.5)	0.119	1.14	-2.479	-
5	sqrt-time t	4.72%	5.66%	6.97%	1.038	14 (8.5)	0.166	1.11	-2.498	0.077
5	daily t path-sum	4.31%	5.04%	5.97%	0.923	15 (8.5)	0.082	1.21	-2.401	0.142

h	Method	Mean ES 95%	Mean ES 97.5%	Mean ES 99%	ES ratio vs sqrt-G 97.5%	Hits 97.5% (exp.)	MF p 97.5%	Loss/ES 97.5%	FZ0 97.5%	DM p vs sqrt-G
5	horizon-matched	4.35%	5.13%	6.13%	0.939	15 (8.5)	0.091	1.19	-2.420	0.167
5	<i>realised mean of worst α share</i>	7.25%	9.85%	14.72%						
20	sqrt-time Gaussian	9.17%	10.85%	13.12%	1.000	3 (2.1)	0.259	1.07	-1.942	-
20	sqrt-time t	9.40%	11.26%	13.86%	1.038	3 (2.1)	0.259	1.04	-1.933	0.228
20	daily t path-sum	7.71%	9.05%	10.70%	0.834	3 (2.1)	0.034	1.36	-1.715	0.454
20	horizon-matched	7.86%	9.30%	11.11%	0.857	3 (2.1)	0.034	1.31	-1.734	0.474
20	<i>realised mean of worst α share</i>	12.14%	15.66%	18.94%						

Table 5. Out-of-sample h -day ES on non-overlapping blocks (339 five-day, 84 twenty-day). ES is a positive loss in percent of portfolio value. The realised row is the average of the worst 5%, 2.5% and 1% of realised h -day losses, for scale only.

Simulating the path lowers 20-day ES by about 17% relative to sqrt-time scaling (14% for the horizon-matched version). There are two reasons. Summing daily t shocks thins the tails (a central-limit effect), and the GARCH variance path mean-reverts, whereas $\sqrt{h}$ scaling keeps today's volatility and today's fat tails for all h days. Using the H2 copula instead of the daily one adds back only 2.3% of ES, consistent with the flat tail dependence of Section 4.

The data side with the more conservative number. At 97.5% the McNeil-Frey test rejects the path-sum ES ($p = 0.034$) but not the sqrt-time ES ($p = 0.259$). Every method sits below the realised tail: the average of the worst 2.5% of 20-day losses was 15.7% against a mean sqrt-time ES of 10.9%. That gap comes from what none of the models contain, namely drawdowns that persist across days and dependence that rises during a crisis (Table 2). Read this way, $\sqrt{h}$ scaling is **conservative by accident**: its overstatement of fat tails offsets missing crisis dynamics. A desk that "improves" the shortcut with an iid path simulation would cut capital by double digits and be less safe. Power is low, however. With 84 twenty-day blocks, only 3 exceedances occur at 97.5% (2.1 expected), and no Diebold-Mariano test separates the methods (all $p > 0.23$).

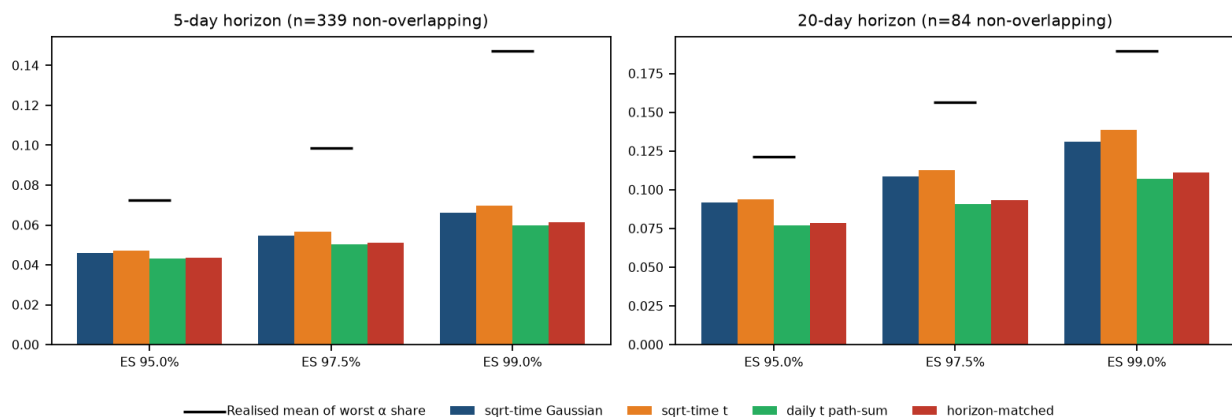


Figure 5. Mean out-of-sample h -day ES by method against the realised average of the worst α share of h -day losses (black bars).

7. How these results compare with the literature

Study	Market, method	Out-of-sample backtest	Model-free tail check	h -day ES	Main scale finding
Jammazi & Reboredo (2016)	Oil-stock, à trous wavelet + copulas	No	No	No	Dependence rose with scale before Lehman and at all scales after
Shahzad et al. (2016)	Greece vs 11 EU markets, wavelet/VMD + time-varying SJC	No	No	No	Long-run tail dependence higher and steadier; short run spikes in crises
Aloui et al. (2013); Aloui & Jammazi (2015)	Agri and oil-FX, wavelet-copula VaR	Partly (VaR)	No	No	Wavelet-filtered models improve VaR/ES against raw
Cai et al. (2020)	Oil vs East Asian equities, MODWT + SJC	No	No	No	Tail dependence insignificant at D1, rises with scale

Study	Market, method	Out-of-sample backtest	Model-free tail check	h-day ES	Main scale finding
Alqaralleh & Canepa (2021)	Six equity markets, WC-GARCH + Clayton	No	No	No	COVID raised tail dependence mainly at 2–8 days
This study	7 US sectors, MODWT-MRA + Gaussian/t/Clayton, GARCH-t filter	Yes: 8 models, Kupiec, Christoffersen, McNeil-Frey, FZ0+DM	Yes: $\lambda_L(q)$ with paired block bootstrap	Yes: 5 and 20 days vs $\sqrt{h}$	Flat in calm periods ($p = 0.31$); long scale jumps in crises; family, not scale, drives ES

Table 6. Positioning against published wavelet-copula studies.

Our in-sample results are in line with the literature in finding substantial tail dependence at every scale, and our crisis split agrees with Jammazi and Reboredo (2016) and Shahzad et al. (2016) that crises lift dependence. We differ in three ways. First, the scale pattern several papers report comes from a fitted copula; with a model-free estimator and honest intervals we cannot reject flatness for a liquid, highly integrated sector basket, and we show that the t copula can manufacture a declining pattern through v alone. Second, the reported VaR gains of wavelet filtering (Aloui et al. 2013; Aloui and Jammazi 2015) do not appear once the margins carry a daily GARCH filter and models are compared with a strictly consistent loss. Third, the literature rarely asks what horizon dependence does to multi-day ES; for this portfolio it is small next to the effect of how the horizon is aggregated.

8. Critical review and limitations

- **Static copulas.** Dependence is re-estimated every 20 days but held fixed in between. Exceedance clustering in March 2020 shows the cost. Time-varying (DCC or score-driven) or regime-switching copulas, or vine copulas with pair-specific tails, are the natural next step.
- **Single-parameter Clayton.** One θ for 21 pairs explains its poor AIC. Its good ES calibration is useful evidence about the tail, not a production recommendation.
- **Low power at long horizons.** 84 non-overlapping 20-day blocks give a few exceedances. Overlapping windows add observations but need HAC-adjusted tests; we chose the honest smaller sample.
- **Pre-asymptotic tail dependence.** $\lambda_L(q)$ at $q = 5\text{--}10\%$ measures joint extremes at practical quantiles, not the mathematical limit $q \rightarrow 0$. It is the quantity that matters for VaR and ES at these levels, but it is not directly comparable with copula limits.
- **Band trimming and long bands.** Trimming the 190-day boundary removes roughly a year of data from each band, and H3 has few effective observations. Crisis-window H3 values are therefore descriptive.
- **Scope.** Liquid US ETFs with equal weights. Thinly traded or emerging markets, where stale prices create genuine horizon effects, may behave differently.
- **Corrections to our first draft.** An internal review of our first version found six errors: a pairwise Clayton likelihood compared against full likelihoods in AIC; a t copula scored with a normal density, so v always hit its upper bound; λ_L read off Kendall's τ rather than measured; wavelet coefficients summed instead of MRA components, with circular boundaries contaminating the latest observations; volatility frozen for 120 days; and 1-day risk only. All six are fixed here. The Clayton-wins result and the 28% ES gap of that draft do not survive.

9. Recommendation for a risk manager

- **Move the daily ES engine from a Gaussian to a Student-t copula.** The t copula wins AIC in 100% of refits, raises 97.5% ES by about 4%, and removes the McNeil-Frey rejection that the Gaussian copula fails at 3 of 3 levels. Track a Clayton-copula ES (+14%) as a challenger number for the lower tail.
- **Do not add horizon-specific (wavelet) copulas to the production engine for this book.** They add complexity with no measurable gain (no Diebold-Mariano p below 0.12), because calm-period tail dependence does not vary with horizon (p = 0.31).
- **Keep $\sqrt{h}$ scaling as the floor for 10-20 day ES,** and do not replace it with an iid path simulation: that would cut 20-day ES by about 17%, and the lower number is the one the backtest rejects (McNeil-Frey p = 0.034).
- **Add a crisis-dependence stress test** that re-prices multi-day ES with long-horizon tail dependence at the crisis level (0.92, against 0.39 in calm periods), because every method under-predicted the realised 20-day tail (15.7% against 10.9%).

10. Reproducibility

From the repository root, make test runs the unit tests (MRA reconstruction, copula densities against an independent library, tail coefficient formulas, Kupiec values, FZ0 propriety) and make reproduce re-runs the whole pipeline and rebuilds this PDF in about 4 minutes on a laptop. The seed is 42, the data are cached, and every number in the text and tables is read from outputs/*.json; none is typed by hand.

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Appendix A. Use of AI tools

AI coding assistants (Cursor agents) were used for literature search support, code scaffolding, debugging, test writing and drafting of this report. The team set the research question, the asset universe, the model design and the validation protocol, reviewed every module, and can run, explain and modify the code. All numerical results are produced by the pipeline and inserted into the text automatically; the AI tools did not supply any number.

Appendix B. Out-of-sample AIC winners and settings

Dependence input	Gaussian	Student-t	Clayton
raw	0%	100%	0%
H1	0%	100%	0%
H2	0%	100%	0%

Table B1. Share of the 85 out-of-sample refits won by each family on AIC.

Setting	Value
Sample	1999-01-05 to 2026-09-30, 6,977 days
Training / out of sample	to 2019-12-31 / from 2020-01-01
Rolling window, refit step	1,000 days, every 20 days
Simulations per refit, seed	20,000, 42
Wavelet, levels	db2, J = 6 (MODWT MRA, reflection, trimmed)
Bootstrap replications	500 (stationary bootstrap)
Pipeline runtime	229 s